

Zeshui Song, 1/15/26, HW1

1.5 Two steel plates are to be held together by means of 16-mm-diameter high-strength steel bolts fitting snugly inside cylindrical brass spacers. Knowing that the average normal stress must not exceed 200 MPa in the bolts and 130 MPa in the spacers, determine the outer diameter of the spacers that yields the most economical and safe design.

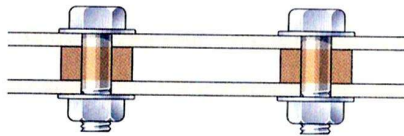
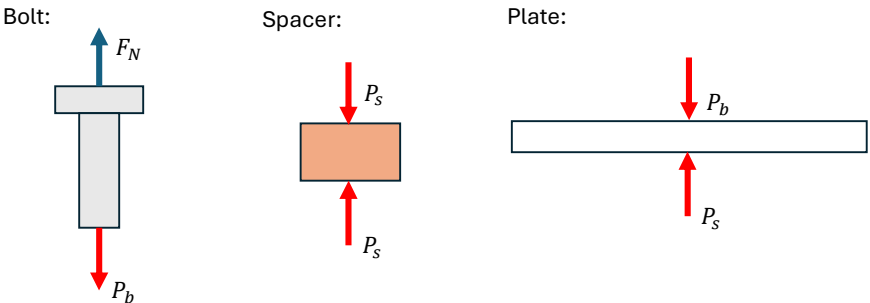


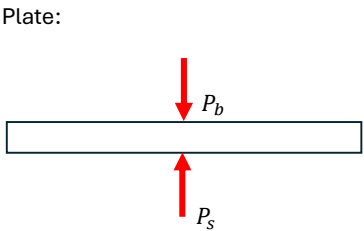
Fig. P1.5

- Given:
- Bolt diameter: 16mm
 - Max average normal stress in bolt: 200 Mpa
 - Max average normal stress in spacers: 130 Mpa
 - Assuming massless components
- Find:
- Outer diameter of spacers.



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- Approach:
- Using the plate FBD, I can use the equation $\sigma = \frac{P}{A}$ to find the max applied load based on the max stress on the bolt and spacer. I can say that the bolt applies P_b to the plate because the bolt is in static equilibrium thus the normal force between the bolt and plate must equal P_b . Finally, using a static equilibrium force balance on the plate, I can solve for the area and diameter of the spacer.

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$$\sum F_y = P_s - P_b = 0 \rightarrow P_s = P_b$$

$$\sigma_s = \frac{P_s}{A_s} = \frac{P_s}{\pi \left(\left(\frac{d_s}{2} \right)^2 - \left(\frac{d_b}{2} \right)^2 \right)} \rightarrow P_s = \frac{1}{4} \pi \sigma_s (d_s^2 - d_b^2)$$

$$\sigma_b = \frac{P_b}{A_b} = \frac{P_b}{\pi \left(\frac{d_b}{2} \right)^2} \rightarrow P_b = \frac{1}{4} \pi \sigma_b d_b^2$$

$$\frac{1}{4} \pi \sigma_b d_b^2 = \frac{1}{4} \pi \sigma_s (d_s^2 - d_b^2)$$

$$\sigma_b d_b^2 = \sigma_s d_s^2 - \sigma_s d_b^2$$

$$(\sigma_b + \sigma_s) d_b^2 = \sigma_s d_s^2$$

$$d_s = d_b \sqrt{\frac{\sigma_b}{\sigma_s} + 1}$$

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$$d_s = d_b \sqrt{\frac{\sigma_b}{\sigma_s} + 1}$$

$$d_s = [16 \text{ mm}] \sqrt{\frac{[200 \text{ MPa}]}{[130 \text{ MPa}]} + 1}$$

$$d_s = 25.49 \text{ mm}$$

Time spent: 1 hr

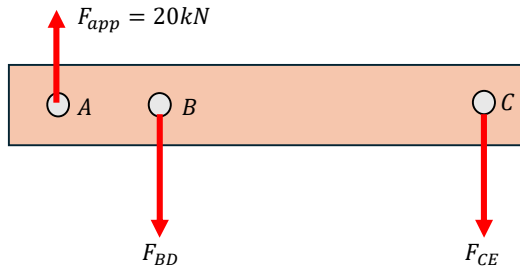
This solution makes sense since the max stress on the spacer is lower than the max stress on the bolt. Thus, it has to have a larger diameter (area) to compensate.

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1.9 Each of the four vertical links has an 8 x 36-mm uniform rectangular cross section and each of the four pins has a 16-mm diameter. Determine the critical values of average normal stress in the links connecting (a) points B and D, (b) points C and E. Hints: (1) consider the loading state-(compression or tension?) in the member, and (2) compute the bearing stresses in all cases.

Assuming members are in tension:



Given:

- Assuming massless components
- The 4 vertical links have a 8 x 36 mm rectangular cross section
- Each of the 4 pins have a 16 mm diameter

Find:

- The critical value of average normal stress in links at BD and CE.
- Compute bearing stress at BD and CE.

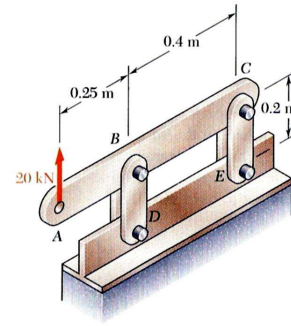


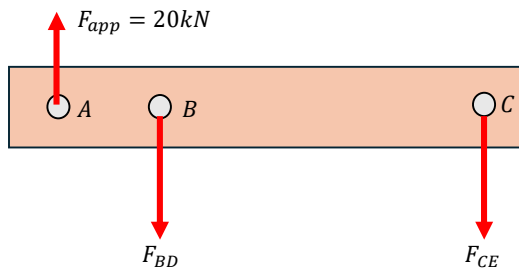
Fig. P1.9

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Approach:

- Using a static equilibrium analysis on member ABC, I can determine the forces on the links at BD and CE. Then, I can solve for the critical stress using the appropriate cross-sectional area: the net area (subtracting the pin hole) for members in tension, or the full area for members in compression to check for buckling in the middle section. Finally to find bearing stress, I can use the same forces but applied to the projected area of the pin (diameter * thickness).



$$\sum M_C = -F_{app}[0.65 \text{ m}] + F_{BD}[0.4 \text{ m}] = 0$$

$$F_{app}[0.65 \text{ m}] = F_{BD}[0.4 \text{ m}] \rightarrow F_{BD} = 1.625F_{app}$$

$$\sum F_y = F_{app} - F_{BD} - F_{CE} = 0$$

$$F_{app} - 1.625F_{app} - F_{CE} = 0 \rightarrow F_{CE} = -0.625F_{app}$$

$$F_{BD} = 1.625F_{app} = 1.625[20 \text{ kN}] = 32.5 \text{ kN}$$

$$F_{CE} = -0.625F_{app} = -0.625[20 \text{ kN}] = -12.5 \text{ kN}$$

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$$F_{BD} = 32.5 \text{ kN}$$

$$F_{CE} = -12.5 \text{ kN}$$

Each of the links at BD and CE carry ½ of the total load since there are 2 links at each point.

Critical stress of a single link at BD:

$$\sigma_{BD} = \frac{\frac{1}{2} F_{BD}}{t(w-d)} \quad \sigma_{BD} = \frac{\frac{1}{2} [32.5 \text{ kN}] \left[\frac{10^3 \text{ N}}{\text{kN}} \right]}{[8 \text{ mm}] \left[\frac{10^{-3} \text{ m}}{\text{mm}} \right] ([36 \text{ mm}] - [16 \text{ mm}]) \left[\frac{10^{-3} \text{ m}}{\text{mm}} \right]} = 101.56 \text{ MPa}$$

Critical stress of a single link at CE:

$$\sigma_{CE} = \frac{\frac{1}{2} F_{CE}}{tw} \quad \sigma_{CE} = \frac{\frac{1}{2} [-12.5 \text{ kN}] \left[\frac{10^3 \text{ N}}{\text{kN}} \right]}{[8 \text{ mm}] \left[\frac{10^{-3} \text{ m}}{\text{mm}} \right] ([36 \text{ mm}] \left[\frac{10^{-3} \text{ m}}{\text{mm}} \right])} = -21.70 \text{ MPa}$$

$$\sigma_{BD} = 101.56 \text{ MPa (Tension)}$$

$$\sigma_{CE} = -21.70 \text{ MPa (Compression)}$$

This solution makes sense since this system is like a lever that pulls up on links BD and pushes down on links CE, Thus BD is in tension and CE is in compression. Also, the effective cross-sectional area for CE is larger since in compression, there are no resulting gaps in the material.

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$$F_{BD} = 32.5 \text{ kN}$$

$$F_{CE} = -12.5 \text{ kN}$$

Each of the links at BD and CE carry ½ of the total load since there are 2 links at each point.

Bearing stress of a pin at BD:

$$\sigma_b = \frac{\frac{1}{2} F_{BD}}{dt} \quad \sigma_b = \frac{\frac{1}{2} [32.5 \text{ kN}] \left[\frac{10^3 \text{ N}}{\text{kN}} \right]}{[8 \text{ mm}] \left[\frac{10^{-3} \text{ m}}{\text{mm}} \right] ([16 \text{ mm}] \left[\frac{10^{-3} \text{ m}}{\text{mm}} \right])} = 126.95 \text{ MPa}$$

Bearing stress of a pin at CE:

$$\sigma_c = \frac{\frac{1}{2} F_{CE}}{dt} \quad \sigma_c = \frac{\frac{1}{2} [-12.5 \text{ kN}] \left[\frac{10^3 \text{ N}}{\text{kN}} \right]}{[8 \text{ mm}] \left[\frac{10^{-3} \text{ m}}{\text{mm}} \right] ([16 \text{ mm}] \left[\frac{10^{-3} \text{ m}}{\text{mm}} \right])} = -48.82 \text{ MPa}$$

$$\sigma_b = 126.95 \text{ MPa}$$

$$\sigma_c = 48.82 \text{ MPa}$$

Time spent: 1.5 hrs

Note: You should not report a negative bearing stress for one member. Bearing stress is always compressive (like a pressure), whether the member is in tension or compression. So we clearly define that we are talking about bearing stress, and report positive numbers.

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1.18 Two wooden planks, each 12 mm thick and 225 mm wide, are joined by the dry mortise joint shown. Knowing that the wood used shears off along its grain when the average shearing stress reaches 8 MPa, determine the magnitude P of the axial load that will cause the joint to fail.

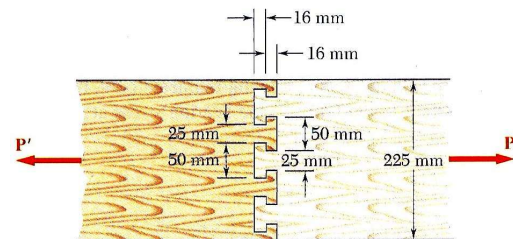


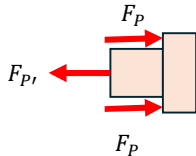
Fig. P1.18

Given:

- Each wooden plank is 12mm thick and 225 mm wide
- The wood shears when avg shear stress reaches 8 MPa

Find:

- Magnitude of P when the joint fails.



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Approach:

- For this joint to fail, the wood must shear off at the 6 places where the darker and lighter wooden plank overlaps. Since we are given the average shear stress, we can find the net area by summing up the 6 cross sectional areas to find the force applied (P).

Single shearing area:

$$lt = 16 * 12 [mm^2] \left[\frac{(10^{-3}m)^2}{mm^2} \right] = 0.192 \times 10^{-3} m^2$$

Net shearing area:

$$6lt = 6 * 16 * 12 [mm^2] \left[\frac{(10^{-3}m)^2}{mm^2} \right] = 1.152 \times 10^{-3} m^2$$

$$\tau = \frac{P}{6lt} \rightarrow P = 6lt\tau$$

$$P = [1.152 \times 10^{-3} m^2][8 MPa] \left[\frac{10^6 Pa}{MPa} \right] = 9.216 kN$$

While shear stress seemed so large, the area it acts over is very small, so it results in a moderate tensile force.

Time spent: 40 mins

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1.30 Two wooden members of uniform rectangular cross section are joined by the simple glued scarf splice shown. Knowing that the maximum allowable shearing stress in the glued splice is 620 kPa, determine (a) the largest load **P** that can be safely applied, (b) the corresponding tensile stress in the splice.

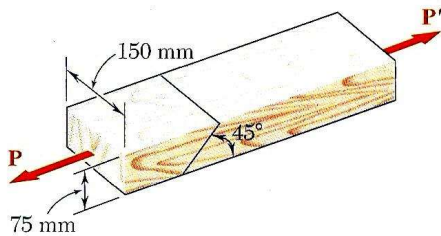


Fig. P1.29 and P1.30

Given:

- Wood : 75x150 mm
- Joined together a 45 degs
- Max shear stress is 620 kPa

Find:

- Magnitude of P when the joint fails.
- Corresponding tensile stress

Approach:

- Using the shear and normal stress equations

$$\sigma = \frac{P}{A_0} \cos^2 \theta$$

$$\tau = \frac{P}{A_0} \sin \theta \cos \theta$$

I can solve for P given the max shear stress. Once P is known, I can find the corresponding tensile stress.

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$$\tau = \frac{P}{A_0} \sin \theta \cos \theta \rightarrow P = \frac{\tau A_0}{\sin \theta \cos \theta}$$

$$P = \frac{[620 \text{ kPa}] \left[\frac{10^3 \text{ Pa}}{\text{kPa}} \right] [75 * 150 \text{ mm}^2] \left[\frac{(10^{-3} \text{ m})^2}{\text{mm}^2} \right]}{\sin(45^\circ) \cos(45^\circ)} = 13.95 \text{ kN}$$

$$P = 13.95 \text{ kN}$$

$$\sigma = \frac{P}{A_0} \cos^2 \theta = \frac{[13.95 \text{ kN}] \left[\frac{10^3 \text{ Pa}}{\text{kPa}} \right]}{[75 * 150 \text{ mm}^2] \left[\frac{(10^{-3} \text{ m})^2}{\text{mm}^2} \right]} \cos^2(45^\circ) = 620 \text{ kPa}$$

$$\sigma = 620 \text{ kPa}$$

This tensile stress makes sense since the oblique plane is at a 45 deg angle which means the shear and tensile stress components should be equal.

Time spent: 30 mins

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